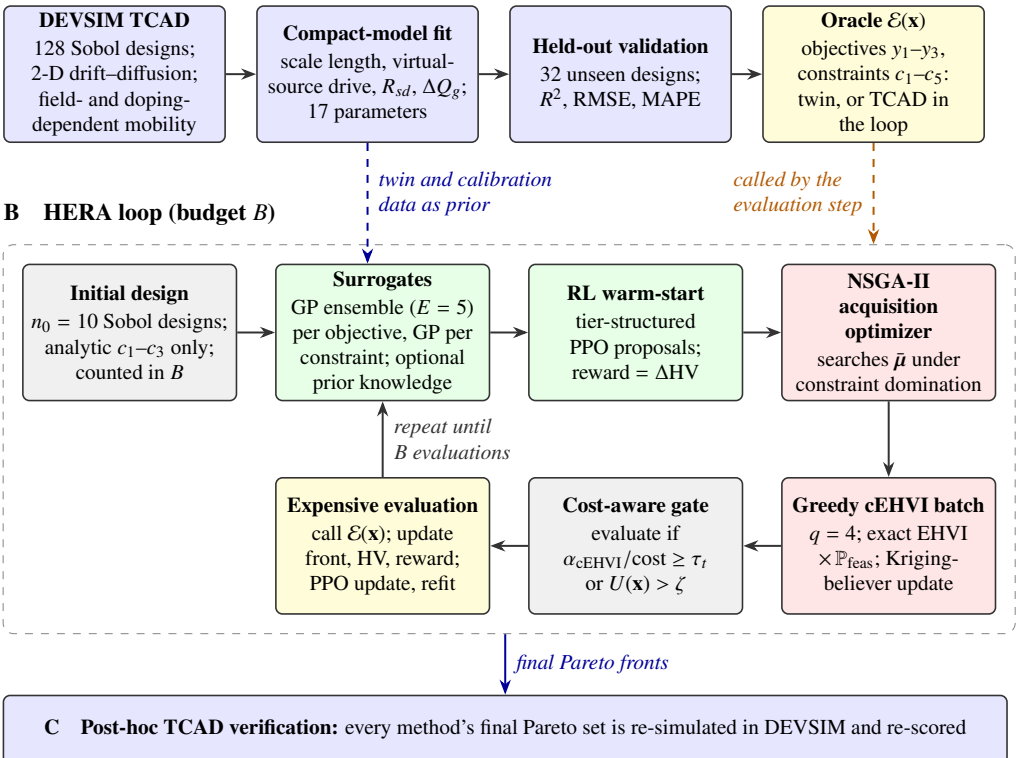

A Offline construction of the TCAD-calibrated digital twin (once)
DEVSIM TCAD
128 Sobol designs; 2-D drift–diffusion; field- and doping-dependent mobility
Compact-model fit
scale length, virtual-source drive, R_{sd}, ΔQ_g; 17 parameters
Held-out validation
32 unseen designs; R^2, RMSE, MAPE
Oracle $\mathcal{E}(\mathbf{x})$
objectives y_1–y_3, constraints c_1–c_5: twin, or TCAD in the loop
twin and calibration data as prior
called by the evaluation step
B HERA loop (budget B)
Initial design
$n_0 = 10$ Sobol designs; analytic c_1–c_3 only; counted in B
Surrogates
GP ensemble ($E = 5$) per objective, GP per constraint; optional prior knowledge
RL warm-start
tier-structured PPO proposals; reward = ΔHV
NSGA-II acquisition optimizer
searches $\tilde{\boldsymbol{\mu}}$ under constraint domination
repeat until B evaluations
Expensive evaluation
call $\mathcal{E}(\mathbf{x})$; update front, HV, reward; PPO update, refit
Cost-aware gate
evaluate if $\alpha_{\mathrm{cEHVI}}/\mathrm{cost} \geq \tau_t$ or $U(\mathbf{x}) > \zeta$
Greedy cEHVI batch
$q = 4$; exact EHVI $\times \mathbb{P}_{\mathrm{feas}}$; Kriging-believer update
final Pareto fronts
C Post-hoc TCAD verification: every method's final Pareto set is re-simulated in DEVSIM and re-scored